Neeraj Sahu

5 YoE as Control Engineer | MTech, IIT Madras

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Skills

Control & Optimization:	Wind Farm Control, Wake Steering (Yaw Offset Control), State-Space Models, Kalman Filtering (KF/EKF/UKF), Uncertainty Quantification
Simulation & Modeling:	BHaWC (Aeroelastic Simulation), FLORIS (Wake Modeling), PyMC (Bayesian Estimation)
Standards & Certification:	IEC 61400 Standards, Controller Type Certification, Verification & Validation (V&V)
Languages:	Python, MATLAB, Bash, R
Tools & Frameworks:	Git, LaTeX, Jira, Linux, SCADA Analytics

Experience

Siemens Energy Wind Power Solution Designer - R&D *July 2021 – Present*

- **Quarterly Controller Certification:** Owned the planning and execution of quarterly certification documentation for the D8, SG5X, and NG-X Controllers, supervising with a senior consultant and intern to meet deadlines.
  - Aligned with multiple stakeholders such as Solution Owners, Product Owners, and Tool Developers.
  - Evangelized a comprehensive understanding of the certification process, including the IEC 61400 standard and its application to wind turbine control systems.
  - Developed a robust framework for the certification process, ensuring that all necessary documentation and testing procedures were in place.
  - Implemented a rigorous testing and validation process to ensure that all control systems met the required standards and specifications.
- **Wind Farm Control:** Developed and implemented advanced control algorithms for wind farm optimization, resulting in improved energy production and reduced maintenance costs.
  - Conducted extensive simulations and modeling to evaluate the performance of control strategies under various operating conditions.
  - Collaborated with cross-functional teams to integrate control systems with existing infrastructure and ensure seamless operation.
- **Performance Validation:** Developed a method to quantify the power benefit of wake steering in offshore wind farms using only existing SCADA data, without the need for external measurement tools.
  - Implemented uncertainty quantification and optimization techniques to improve the accuracy and reliability of the wake steering power benefit measurements.

Industry Mention @ IEA Wind Task 44

An industry-oriented method to measure Wake Steering Power Benefit in Offshore wind farms

Click here for YouTube presentation.

- Our work was invited for a presentation at the IEA Wind Task 44 Workshop, where we shared our industry-oriented method to measure wake steering power benefit in offshore wind farms.

Education

M.Tech. in Aerospace Engineering *2019 – 2021*

Indian Institute of Technology, Madras *Chennai, India*

- **Master's Thesis:** *Statistical Estimation of Motion Parameters from Radar Signals of Manned Maneuvering Targets.*
- Formulated and benchmarked Linear Kalman (KF), Extended Kalman (EKF), and Unscented Kalman Filter (UKF) algorithms for nonlinear state estimation and trajectory tracking in noisy radar data.
- Applied statistical estimation, time-series analysis, optimization, and nonlinear dynamic modeling in MATLAB and Python.
- **Relevant Coursework:** Data Science & Analytics, Numerical Optimization, Control Systems.

BEng in Automobile Engineering *2015 – 2019*
Madhav Institute of Technology and Science (MITS) *Gwalior, India*

- Built a strong foundation in Mechanical Engineering, Vehicle Dynamics, Control Systems, Mechatronics, and Mathematical Modeling.

Professional Development

Technical Project Management *Mar 2026*
Critical Conversations and Conflict Management *Aug 2025*